iC-TW28 Technical Digest

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1. Chip identity

Item	Value
Full part name	**iC-TW28** — "10-BIT SIN/COS INTERPOLATOR WITH AUTO-CALIBRATION AND LINE DRIVER"
Manufacturer	**iC-Haus GmbH**, Am Kuemmerling 18, D-55294 Bodenheim, Germany — www.ichaus.com
Data sheet	Rev **D5**, 2022-11-24
Package / order code	**QFN32, 5 mm × 5 mm × 0.9 mm**, RoHS compliant; order designation `iC-TW28 QFN32-5x5`
Supply	Single **3.3 V** (3.1 … 3.6 V), three pin pairs AVDD/AVSS, DVDD/DVSS, IOVDD/IOVSS
Device ID register	`DEVICE_ID` (0xE000) = **0x0C**
Revision registers	`REV1` / `REV0` (0xE003/0xE002): 0x11/0x14 = iC-TW28 X, 0x11/0x15 = iC-TW28 W and W1, 0x04/0x17 = iC-TW28 V1
Evaluation kit	`iC-TW28 EVAL TW28_1D` PCB (≈68 mm × 102 mm) + free Windows GUI

Function (one paragraph)

The iC-TW28 is a general-purpose **10-bit application-specific interpolator** for sine/cosine sensor signals. It accepts differential analog sin/cos signals (and an optional differential index/zero signal) directly from magnetic or optical sensors, amplifies them with programmable-gain amplifiers, applies coarse **analog** offset and gain correction, digitizes them with 10-bit ADCs (8-bit for the zero channel), then applies **digital** offset/amplitude-balance/phase correction, computes the angle within one input cycle as `arctan(Sin/Cos)` with a **CORDIC** algorithm at **10-bit resolution (1024 increments per input cycle)**, filters the result in a configurable PI servo loop, scales it to a programmable **interpolation factor** and emits industry-standard **ABZ incremental quadrature** and/or **UVW block-commutation** outputs through a built-in **RS422-compatible differential line driver** (or as CMOS single-ended outputs). A 14-bit multi-cycle counter plus a capture register provide part of an absolute-position path over the SPI port. Configuration and calibration are held in an on-chip EEPROM and can be programmed by pins, by SPI, or in the field through the two-wire **Encoder Link** interface.

Target applications (as listed)

- Rotary and linear encoders
- Magnetic or optical sin/cos sensor interface
- Brushless motor commutation (**2 … 64 poles**; description text also says "1 to 32 pole-pair motors")
- Embedded motion control

Key selling points / headline numbers

Parameter	Value
Angle resolution	**10 bits = 1024 increments per input cycle**
Interpolation factor	**2 … 256** (serial configuration mode); **24 factors** in pin configuration mode (Table 2, groups I0/I1; includes 1 … 256)
Max sin/cos input frequency	**700 kHz** (elec. char. item 104); higher allowed in SPI-Only mode because the AB output limit then does not apply
AB output frequency	up to **12.5 MHz** (features + ABLIMIT formula); description text says "up to 12 MHz (20 ns edge spacing)"
Minimum AB edge spacing	**20 ns** nominal (`tMTD` = 1/f_osc at ABLIMIT = 0x00)
Latency	**typ. 1.5 µs** (`MAIN_CFG.noise = 0`), **1.9 µs** (`MAIN_CFG.noise = 1`)
INL / DNL	0.7 ° / 0.35 ° (1 Vpp-diff input, compensated offset/gain/phase)
Multi-cycle counter	**14 bits**, up to **16,383 input cycles**
Position value	**24 bits** = 14-bit multi-cycle count + 10-bit angle
Internal oscillator	~50 MHz (48…51 MHz @3.1 V, 49…52 MHz @3.6 V), TC 225 ppm/K
Auto functions	Auto-calibration (offset, gain/balance, phase, Z channel) and continuous **auto-adaption**
Outputs	Differential RS422 ABZ *or* UVW, **and** simultaneous single-ended ABZ + UVW (ABZUVW mode); SPI-Only mode
Configuration	Pin configuration (8 resistors) or serial (SPI or Encoder Link); EEPROM stored
Extras	LED intensity control PWM output, capture register for coded reference marks / touch-probe, fault/IRQ output, watchdog

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2. Pinout / interfaces

QFN32-5x5, top view. Backside paddle = TP2 (must be connected to a ground plane at AVSS potential; may also be used to connect DVSS).

Pin	Name	Type	Function / notes
1	SIN+	Analog in	Differential sine input +

Pin	Name	Type	Function / notes
2	SIN–	Analog in	Differential sine input –
3	AVDD	Supply	Analog +3.1…3.6 V; tie to DVDD, IOVDD from a clean source
4	COS+	Analog in	Differential cosine input +
5	COS–	Analog in	Differential cosine input –
6	AVSS	Ground	Analog ground, solid plane
7	ZERO+	Analog in	Differential zero/index input +
8	ZERO–	Analog in	Differential zero/index input –
9	VREF	Analog out	ADC reference output. Decouple **100 nF** to AVSS. No noise injection
10	VC	Analog out	Bias output = **VDD/2**. Decouple **100 nF** to AVSS. Must **not** be used to bias external circuitry or single-ended sin/cos inputs
11, 12, 13	Reserved	Digital in	Must be tied to **ground** (iC-TW29 uses pin 13 as GPIO)
14	xRST	Digital in, active low	Reset; device held reset while low. RC 1 kΩ + 100 nF recommended
15	xCALIB	Digital in, active low	Auto-calibration on falling edge; tie high if unused
16	xIRQ	Digital out, active low	Interrupt / fault output; push-pull or open-drain. Can drive a fault LED
17	Z–	Digital / RS422 out	ABZ mode: Z–. UVW mode: W–
18	Z+	Digital / RS422 out	ABZ mode: Z+. UVW mode: W+
19	IOVSS	Ground	I/O ground
20	B–	Digital / RS422 out	ABZ mode: B–. UVW mode: V–. Z-calib mode: Z gating window (inverted)
21	B+	Digital / RS422 out	ABZ mode: B+. UVW mode: V+. Z-calib mode: Z gating window
22	IOVDD	Supply	Output driver supply (incl. RS422 and LED); must equal DVDD; can draw up to 100 mA

Pin	Name	Type	Function / notes
23	A–	Digital / RS422 out	ABZ mode: A–. UVW mode: U–. Z-calib mode: un-gated Z (1/input period). Encoder Link: **ELIN input / ELOUT output**
24	A+	Digital / RS422 out	ABZ mode: A+. UVW mode: U+. Z-calib mode: un-gated Z. Encoder Link: **ELCLK input**
25	DVDD	Supply	Digital +3.1…3.6 V
26	LED	Digital out	LED PWM / intensity-control output; push-pull or open-drain
27	DVSS	Ground	Digital ground
28	SO/C3	Digital out / analog in	SPI slave output (to master MI) or pin-config input **C3**
29	SI/C2	Digital in / analog in	SPI slave input (from master MO) or pin-config input **C2**
30	SCLK/C1	Digital in / analog in	SPI clock input or pin-config input **C1**
31	xSS/C0	Digital in / analog in	SPI slave select (active low) or pin-config input **C0**
32	PINCFG	Digital in	**DVDD = pin configuration mode**, **DVSS = serial (SPI) configuration mode**. Footnote 6: iC-TW29 uses pin 32 as BISSEN (high selects BiSS) — *this pin is not BiSS on the iC-TW28*

Unused-pin rules (datasheet footnotes): differential/SPI pins → 10 kΩ to ground (SIN/COS/ZERO/SPI); xRST, xCALIB, xIRQ, LED → 3.3 V via 10 kΩ; PINCFG must be tied high or low, never float; reserved pins to ground.

Signal-type / level summary

- **Analog differential**: SIN±, COS±, ZERO± (see §3 for amplitudes and common mode).
- **RS422 differential outputs** (when `MAIN_CFG.rs422 = 1`): A±/U±, B±/V±, Z±/W±, nominal 18…27 mA into a 120 Ω terminated pair; short-circuit currents –55 mA / +35 mA (spec items 902/903; note release W may exceed these at temperature extremes).
- **CMOS digital outputs** (when `MAIN_CFG.rs422 = 0`): same pins, ±10 mA per pin, V_OL / V_OH ≈ 0.7 V at ±4 mA, t_r/t_f = 20 ns into 50 pF.
- **CMOS digital inputs**: V_t(hi) = 1.9 V, V_t(lo) = 0.8 V at DVDD = 3.3 V; leakage ±50 nA.
- **xIRQ / LED**: configurable push-pull or open-drain; LED up to ±15 mA continuous.
- **Encoder Link** (only in serial configuration mode) reuses A+/A– as a 2-wire SPI-like interface.

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3. Analog input front end

Signal levels (elec. char. items 101–111, 201–206; Tables 114, 115)

Input range	Differential amplitude (peak, `max(SIN+−SIN−)`)	Per-input peak	Permissible absolute input voltage	Sensing-input differential R
Low (`MAIN_CFG.input = 0`; C3 level 0/3/6/9)	**20 … 700 mV**	10 … 350 mV	0.35 V … AVDD − 1.1 V	10 MΩ … 1000 MΩ
Low with loss detect (`MAIN_CFG.input = 1`)	20 … 700 mV	10 … 350 mV	as above	**0.240 MΩ** (spec item 111) / **220 kΩ** (Table 114) — the two sources disagree
High (`MAIN_CFG.input = 2`; C3 level 1/4/7/10)	**65 mV … 2000 mV**	32.5 mV … 1.0 V	0 V … AVDD	**0.670 MΩ** (item 111) / **620 kΩ** (Table 114)

- **Common mode** `(SIN+ + SIN−)/2`, likewise for COS: minimum gain: **0.7 V … AVDD − 1.45 V**; maximum gain: **0.35 V … AVDD − 1.1 V**.
- Datasheet warning: "1 Vpp signals with 2.5 V common mode cannot be connected directly to the iC-TW28 as the common mode voltage is too high."
- Amplifier input offset voltage: **±20 mV** (summed with sensor offset for correction). Input leakage ±50 nA.
- ZERO inputs: permissible input 0 … AVDD, common mode as above, V_os ±20 mV, R_in(diff) 10 MΩ … 1000 MΩ.

Internal signal conditioning (in order)

1. **Three programmable gain amplifiers (PGAs)** — one each for sin, cos and zero. The sin/cos PGAs additionally provide a configurable **9 dB attenuation** so rail-to-rail sensor inputs (GMR/TMR, optical) can be used. High input range is implemented with a **resistive attenuator before the PGA**, which is why the input impedance differs between the two ranges.
2. **Noise filter after each PGA** — reduces HF noise from the sin/cos inputs; **two settings** in serial mode.
3. **Analog error correction** — coarse analog offset and **gain correction range 6 … 40.5 dB in 1.5 dB steps** applied *before* A/D conversion.
4. **ADCs** — **10-bit** for sin and cos, **8-bit** for the zero channel. "The remainder of the signal path is completely digital." *Extraction quirk:* the `S_ADC` (0x021A) and `C_ADC` (0x021C) register descriptions say each "contains the MSB of the **12-bit** sin/cos channel ADC value", while the block-diagram text says 10-bit ADCs. The SPI-read *corrected* sin/cos values are given as signed 10-bit; the internal ADC width beyond 10 bits is **not determinable from text**.
5. **Zero channel**: 8-bit ADC + configurable comparator generate the Z gating window (see §4/§6).

External component recommendations (from the electrical-connection figures)

Item	Value
VREF decoupling	100 nF to AVSS (dedicated)
VC decoupling	100 nF to AVSS (dedicated)
Each supply pin (AVDD, DVDD, IOVDD)	1 µF decoupling close to the device
RS422 line termination	**120 Ω at the far (receiving) end** of each of the three pairs; do **not** fit termination when the line driver is disabled
xRST network (optional, recommended)	1 kΩ + 100 nF → ≈**100 µs** delay; larger R gives proportionally longer delay
Pin-configuration resistors	8 resistors (Rx1/Rx2 divider per config pin) to VDD; E48 2 % or E96 1 %; **no decoupling capacitors**; place close to the pins
SPI pull-ups/downs	Required if the host can be disconnected (PINCFG = DVSS)
Single-ended sensor bias	Biased divider to the sensor zero level (typically half sensor supply). Datasheet: do **not** use VC for this purpose because the input attenuator draws too much current
Digital index sensor with open-drain active-low output	Connect to ZERO–, tie ZERO+ to VC (reverse for active-high/open-source)
1 Z pulse per input cycle without index sensor	ZERO+ = 3.3 V, ZERO– = ground (gate window then always high, 360°)
No Z output required	ZERO+ = ground, ZERO– = 3.3 V
Sensor loss detection	Internal high-value resistors bridge the + and – inputs (selected by `MAIN_CFG.input = 1` / `= 2`); floating inputs collapse the amplitude → `STAT_VAL.scamp`

No external signal conditioning is required in most applications ("The iC-TW28 requires no external components for operation" — EEPROM and RS422 drivers are on-chip).

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4. Full signal-processing chain (in order)

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SIN± / COS± ─► PGA (+optional 9 dB atten) ─► noise filter ─► analog offset+gain corr (6…40.5 dB, 1.5 dB
            ─► 10-bit ADC ─► digital correction (offset, balance/gain, phase)
            ─► arctan(Sin/Cos) CORDIC, 10-bit ─► PI filter (kp/ki, optional 640 ns feedback delay)
            ─► hysteresis ─► interpolation scaling (INTER) ─► post-AB divider (1…8)
            ─► ABZ generator (+Z gating path) ─► RS422 / CMOS outputs
ZERO± ─► PGA ─► analog offset/gain corr ─► 8-bit ADC ─► comparator (ZERO0.threshold) ─► Z gating windo
```

4.1 Offset / gain / phase correction and how it is calibrated

Correction	Register(s)	Range / step (datasheet)
Sin analog offset	`S_OFSA_COR` 0x4013	±31 (analog), part of the 6 … 40.5 dB analog gain/offset block
Cos analog offset	`C_OFSA_COR` 0x4015	±31
Sin digital offset	`S_OFS_COR` 0x4012	±127
Cos digital offset	`C_OFS_COR` 0x4014	±127
S/C amplitude balance	`SC_BAL_COR` 0x4016	±127; correctable mismatch **±25 %, step 0.2 %**
S/C digital gain	`SC_GN_COR` 0x4017	0…255
S/C analog gain	`SC_GNA_COR` 0x4018	0…23 (6 … 40.5 dB in 1.5 dB steps)
S/C phase (quadrature)	`SC_PH_COR` 0x4019	±127; correctable phase error **±26 °, step 0.22 °**
Z analog offset	`Z_OFSA_COR` 0x401A	±31
Z analog gain	`Z_GNA_COR` 0x401B	0…23
Correctable total input offset	(auto cal)	**±100 %** of input signal amplitude, step **3.9 mV / gain**
Base values (for adaption limit check)	`S_OFS_BASE` 0x4007, `C_OFS_BASE` 0x4008, `SC_BAL_BASE` 0x400B, `SC_PH_BASE` 0x400E	signed 2's complement
Adaption limits	`SC_OFS_LIM` 0x4009, `SC_BAL_LIM` 0x400C, `SC_PH_LIM` 0x400F	0…127
Residue thresholds	`SC_OFS_TH` 0x400A, `SC_BAL_TH` 0x400D, `SC_PH_TH` 0x4010, `Z_PH_TH` 0x4011	0…127
Residues (read-only)	`S_OFS_RES` 0x4023, `C_OFS_RES` 0x4024, `SC_BAL_RES` 0x4025, `SC_PH_RES` 0x4026, `Z_PH_RES` 0x4028	signed ±127

Calibration scheme - `SC_AMP_TARG` (0x4004) sets the target `√(sin²+cos²)` after A/D — **recommended 150**, allowed 0…180 (180 avoids ADC overload; reported as 100 % in the GUI). During auto calibration the chip adjusts analog and digital gain to hit this level. - **Hardware auto-calibration**: pull `xCALIB` low → supply sensor motion (a few hundred sin/cos cycles; frequency need not be constant and direction may reverse) → release `xCALIB`. In pin configuration mode release stores the values to EEPROM. In serial mode storage on release is controlled by `ADAPT_CFG1.xcalee`; if `xcalee = 0`, issue COMMAND 0x12 (copy COR → BASE) then 0x11 (write to EEPROM). - **Software auto-calibration** (serial mode only): COMMAND **0x23** (all parameters), **0x21** (sin/cos only), **0x22** (Z channel only); terminate with **0x20**; then 0x12 then 0x11. - In pin configuration mode `xCALIB` can be tied permanently to ground → the device self-calibrates at **every startup** (no production calibration step). - **Auto-adaption** runs during operation, controlled per-channel by `ADAPT_CFG0` bits (`scofs`, `scbal`, `scph`, `scofsa`, `scgain`, `zofs`, `zgain`, `zphase`; recommended `ADAPT_CFG0 ≤ 7` so that analog offset/gain and all Z adaption are used for initial calibration only) and by `ADAPT_CFG1.p` (adaption rate: 1/32, 1/16, 1/8, 1/4, 1/2, or 1 COR increment per sin/cos input period), `.stop` (stop adaption

when `STAT_VAL.scamp = 1`), `.zcal`, `.xcalee`. - Calibration target for the Z channel: the ZERO signal is tuned to span **75 % of the 8-bit Z ADC range**, i.e. digitized amplitude **96**; `zphase` is tuned so the Z pulse is **centred** in the gating window (t1 = t2). - Application hint: with no cyclic input, auto calibration only roughly sets gain/offset to avoid ADC overload. - After auto calibration all RES registers should read zero/near zero; if not, repeat calibration.

4.2 Interpolation core

- Algorithm: `arctan(Sin/Cos)` **using a CORDIC algorithm**, resolution **10 bits** (0…1023 per input cycle).
- No arctan lookup table is mentioned; no CORDIC iteration count / pipeline depth is given → **not determinable from text**.
- The 10-bit angle value is scaled to the required output resolution by the interpolation register:
- `INTER0.interlsb` (0x0101) = INTER[7:0], `INTER1.msbs` (0x0102 bits 1:0) = INTER[9:8]
- **`INTER = 256 × INTER1.msbs + INTER0.interlsb`**
- INTER = 0 → **inter = 256**; INTER = 1…7 reserved; INTER = 8…1023 → **inter = INTER/4**
- `inter` = interpolation factor in **AB output cycles per sin/cos input cycle**.
- Output update rate / latency: filter+path latency **1.5 µs** (`MAIN_CFG.noise = 0`) or **1.9 µs** (`MAIN_CFG.noise = 1`). Internal angle/ADC update intervals are 320 ns (see §5). The feature list gives "Low latency (typ. 1.5 µs)".
- Application hint: the chip **must be restarted** after changing `inter` to guarantee proper Z alignment.

4.3 Filter (digital path)

- Configurable **PI servo loop** with optional **640 ns feedback-path delay** (Figure 39).
- `FILT_K.kp` 0…6 (P gain — larger value = more filtering, less noise/jitter); `FILT_K.ki` 0…3 (I gain, **recommended 3**; non-zero ki gives zero lag at constant velocity, smaller ki = less lag under acceleration but possible overshoot on deceleration); `FILT_CFG.fb` 0/1 (640 ns delay; reduces constant-velocity lag by **40 %** with `MAIN_CFG.noise = 0` and **22 %** with `= 1`; requires `FILT_K.kp ≥ 3` for stability); `FILT_CFG.auto` 0/1 (static vs dynamic kp); `FILT_CFG.kpmax` 3…6; `FILT_LAG.threshold` 1…31.
- Instability constraints stated: `FILT_K.kp = 0` with `FILT_K.ki = 1` is unstable; if `FILT_K.ki = 0` then `FILT_K.kp` must also be 0.
- Recommended presets: fastest = kp 0 / ki 3 / fb 0 / auto 0; light = kp 2, medium = kp 4, heavy = kp 6 (all ki 3, fb 0, auto 0).
- Pin mode selects light/medium/heavy filtering through input C1.

4.4 Hysteresis

- `AB.hyst` (0x0103 bits 4:0) 0…31 and `UVW.hyst` (0x0104 bits 4:0) 0…31: **hyst[°] = ± hyst_code × 360° / 2048**. Serial mode = 32 settings; pin mode = 4 settings per path (±0.35°, ±0.7°, ±1.4°, ±2.8°).

4.5 Error / limit monitoring and fault logic

Eight monitored conditions, each available in `STAT_VAL` (live), `STAT_LATCH` (latched, write-0-to-clear), selectable into an interrupt through `STAT_SEL` + `STAT_IE`, and independently able to force the ABZ/UVW outputs to Hi-Z through `STAT_HIZ`:

Bit	Name	Condition
0	`oflow`	Signal path overflow / saturation (likely ADC overflow) — not fatal, degrades accuracy
1	`falarm`	Input frequency above the `FALARM` level; **`f_input[MHz] = 1.56 × FALARM / 256`**, FALARM = 1…128 legal (0 and 129…255 reserved); intended as a high-frequency alarm, not accurate measurement
2	`laglim`	AB outputs lag the input by more than **45 °** of an input cycle (filter is disabled, outputs still valid)
3	`inclim`	AB output frequency is being limited by `ABLIMIT`, **or** the UVW output frequency > **8.33 MHz**; not fatal, outputs still valid
4	`lagfatl`	**Fatal** lag: lag reached **180 °** (half an input cycle), outputs invalid (unexpected direction reversal)
5	`scamp`	`√(sin²+cos²)` outside `SC_AMP_LOW` … `SC_AMP_HIGH`. Pin mode: **< 60 % or > 120 %** of the calibrated value
6	`adapt`	A COR value deviated from its BASE value by more than the corresponding LIM value
7	`res`	A residue exceeded its threshold: `|S_OFS_RES|>SC_OFS_TH`, `|C_OFS_RES|>SC_OFS_TH`, `|SC_BAL_RES|>SC_BAL_TH`, `|SC_PH_RES|>SC_PH_TH`, `|Z_PH_RES|>Z_PH_TH`

- `STAT_CFG.filter` (0…4): condition must persist 0 / 10 µs / 150 µs / 2.5 ms / 40 ms before `STAT_VAL` sets.
- `STAT_CFG.long`: xIRQ prolonged by 40 ms (useful for fault LEDs). `STAT_CFG.enc` = IRQ on multi-cycle-counter overflow; `STAT_CFG.enz` = IRQ on position capture.
- `STAT_FATAL` (read-only): bit0 `eechk` EEPROM checksum error, bit1 `ee2bit` EEPROM double-bit error, bit2 `interr` internal error. Any of these **inhibit startup or stop the device, disable ABZ/UVW, assert xIRQ**, and require a power cycle (or xRST / restart command in hosted systems).
- Pin-configuration-mode fault logic: `STAT_SEL = 0x00` (STAT_VAL used), faults enabled for `scamp`, `lagfatl`, `laglim`; `STAT_HIZ = 0x00`, so only a fatal condition disables the outputs.
- AB output frequency limiter: guarantees a minimum separation between AB edges; **6 choices in pin mode, 256 choices in serial mode**. When limiting, AB lags the input; if transient it catches up, if persistent it eventually becomes a fatal lag fault and the device stops.
- Watchdog: `WATCHDOG` (0x401C) — write 0, wait ≥ 1 ms, should read **31 (0x1F)**; any other value indicates internal malfunction.

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5. Measured angle / position data path

Item	Value
Fine (angle) resolution	**10 bits**, value 0…1023 (0…360 ° of one sin/cos input cycle)
Coarse (multi-turn) resolution	**14 bits** multi-cycle counter, 0…**16,383** input cycles
Combined position	**24 bits** = 14-bit multi-cycle count + 10-bit angle, always read together
Data format	Unsigned binary, MSB first on the wire
Internal update rate	**320 ns** when ABZ outputs are enabled (`OUTPUT.abzen = 1`); **20 ns** when ABZ outputs are disabled (`abzen = 0`, i.e. SPI-Only mode) — the 20 ns value is only *correct* if `INTER = 0` (inter = 256)
Latency	Path latency typ. **1.5 µs** (noise filter setting 0), 1.9 µs (setting 1)
SPI read latency	Overlapped packets: the response to command *n* is sampled on the falling edge of xSS at the start of command *n+1* → **communication latency = one 32-bit SPI command**, not two. At maximum SPI clock a new position can be read **every 2 µs**
Angle always full resolution	"Position (angle) is always returned at the full 10-bit resolution …, regardless of the interpolation value (INTER)"
Captured position	Captured on the **rising edge of the Z output** or on the **Z gating window**, selected by `ZERO0.mode`. Captured values are only valid if `INTER = 0`
ADC read bank	Corrected sin/cos/zero ADC values, updated every **320 ns**

How coarse + fine information is formed

- The 10-bit arctan angle gives the **fine** position inside one input cycle; wrap-around of this angle increments/decrements the **14-bit multi-cycle counter** (the coarse/turn count). Each input cycle = one sin/cos period, so 24-bit position = turns × 1024 + angle.
- The counter can be **cleared on the Z output** (`ZERO0.clr = 1`) or never cleared (`= 0`); it can be **preset over SPI** (Multi-Cycle Counter Write, wm = 1, or atomic read/write, wm = 2) to align with an external absolute system.
- `mcs` **synchronization bit**: states the expected half-period sector of the sensor angle when the counter is preset (0 = 0…180 °/0…511, 1 = 180…360 °/512…1023). If the actual sector has moved, the counter is preset to command ± 1 (specifically −1 if mcs = 0 and 270 ° (768) ≤ angle < 360 ° (1023); +1 if mcs = 1 and 0 ° ≤ angle ≤ 90 ° (256)). This tolerates up to ±90 ° of misalignment or movement during presetting.
- **Rollover flags**: `mcr` is set when the counter passes a multiple of **4,096** cycles (4,096 / 8,192 / 12,288 / 0), with 4,096-cycle hysteresis to avoid repeats on direction reversal; `mcrl` if a second rollover happens before the position is read. Both reset when the position is read. `STAT_CFG.enc` can turn rollover into an xIRQ so a host can extend the counter to any length.
- UVW commutation is derived from the same angle path with its own pole-pair setting and phase shift (§6.2).

6. Output interfaces in detail

Output mode selection (`OUTPUT.uvwen`, `OUTPUT.abzen`, `MAIN_CFG.rs422`):

uvwen	abzen	rs422	Mode	Format	Driver
0	0	X	SPI Only	—	Hi-Z
0	1	0	ABZ	Differential	CMOS
0	1	1	ABZLD	Differential	RS422
1	0	0	UVW	Differential	CMOS
1	0	1	UVWLD	Differential	RS422
1	1	X	ABZUVW	Single-ended (ABZ + UVW)	CMOS

6.1 ABZ incremental output

- **Max output frequency: 12.5 MHz**; **minimum AB edge separation 20 ns** nominal (`tMTD` = 1/f_osc with `ABLIMIT = 0x00`). `fab[MHz] = 12.5 MHz / ((ABLIMIT + 1) × div)` and `t_edge[ns] = 20 × (ABLIMIT + 1) × div`. `ABLIMIT` (0x0106) = 0…255 → max to min output frequency limit (serial mode 256 choices, pin mode 6). The limiter applies to the AB frequency **before** the post-AB divider. Datasheet note: 12.5 MHz AB frequency with 4 edges per cycle ⇒ 20 ns per edge.
- **Pin-configuration-mode frequency limit choices** (C2, Table 4): 195 kHz / 1.28 µs, 390 kHz / 640 ns, 781 kHz / 320 ns, 1.56 MHz / 160 ns, 3.12 MHz / 80 ns, **6.25 MHz / 40 ns** (this is the highest in pin mode), with auto-adaption either enabled (levels 11…6) or disabled (levels 5…0).
- **Programmable resolution / divider**
- `INTER` (10-bit, 0x0101 + 0x0102 bits 1:0): inter = 256 when INTER = 0, else inter = INTER/4 (2… 255.75).
- **Post-AB divider** `INTER1.div` (0x0102 bits 4:2): `div = INTER1.div + 1` → **1…8**; `inter_eff = inter / div`. Gives fractional/non-integer output resolutions (e.g. Table 122: inter_eff 0.25 / 0.50 / 0.75 / 1.00 / 1.25 / 1.50 / 1.75 by combinations of inter and div; worked example: 24 sin/cos cycles/rev → 1024 CPR needs inter = 128 with div = 3 ⇒ 512 CPR).
- **Pin-configuration interpolation factors** (C0/C3, Table 2, "AB output cycles per sin/cos input cycle"; there are 4× as many AB edges): group **I0**: 256, 128, 64, 32, 16, 12, 8, 6, 4, 3, 2, 1; group **I1**: 250, 200, 180, 125, 100, 80, 50, 40, 25, 20, 10, 5.
- **Phase / symmetry specs**: output phase A vs B = **25 %** (item 904, Figure 3 definition), duty cycle at A/B = **50 %** (item 905). Relative angle accuracy `AArel` (ideal 1 kHz waveform, `FILT_K.kp = 6`): **±5 % for INTER < x50, ±10 % for x50…x100, ±15 % for > x100** of the output period.
- **Index (Z) generation rules**
- Z comes from the ZERO channel: 8-bit ADC + comparator; the internal gating window is active while the conditioned ZERO input exceeds **`zthreshold = 4 × ZERO0.threshold`**; threshold in percent of ZERO amplitude = `4 × ZERO0.threshold / 96 × 100 %`. Starting values `ZERO0.threshold = 10`, `ZERO1.zwidth = 0` give zthreshold = 40 → **42 %** of the ZERO signal (these are the fixed pin-mode values).

- Z pulse width: `zwidth[edges] = (ZERO1.zwidth + 1) / div`; `ZERO1.zwidth` (0x0108 bits 3:0) = 0…15 and **must be less than INTER**.
- Window width rule to guarantee exactly one Z per revolution: `zwidth[°] < gatewidth[°] < 720° − zwidth[°]`, with `zwidth[°] = 90° / inter × zwidth[edges]`.
- Z phase: `ZPH = 4 × ZPHASE.msb + PHASE_LSB.z`, `zphase[°] = ZPH × 360° / 1024`. Auto Z calibration centres the pulse in the window (t1 = t2). Manual setting is needed when no index sensor is used (ZERO+ = 3.3 V, ZERO– = GND, gatewidth = 360°).
- The Z output **bypasses the post-AB divider**, so Z-to-AB synchronization may be lost when div ≠ 1.
- Z test mode (`TEST.z = 1`, requires `OUTPUT.abzen = 1`): A outputs = un-gated Z (once per input period), B outputs = internal Z gating window, Z outputs = gated Z pulse.
- `ZERO0.mode` (0x0107 bit 7): 0 = position captured on Z output, 1 = captured on Z gating window.
- `ZERO0.clr` (0x0107 bit 6): 1 = clear the multi-cycle counter on the Z output.
- **Polarity / inversion options**
- `OUTPUT.apol` (bit 0), `.bpol` (bit 1): normal/inverted A and B; these also define the A/B state while Z is active. Setting exactly one of apol/bpol reverses the AB counting direction — then use `abzdir` to restore it. Setting both does not change direction.
- `OUTPUT.zpol` (bit 2): 0 = Z active high, 1 = Z active low.
- `OUTPUT.abzdir` (bit 3): normal/reversed AB counting direction.
- `OUTPUT.abzen` (bit 6): ABZ outputs enable; also selects the 320 ns vs 20 ns internal position update rate.
- Restart required after changing `.apol` / `.bpol` for correct Z alignment.
- **Required external line driver: none.** A **built-in RS422-compatible differential line driver** drives 120 Ω terminated cables (18…27 mA nominal; three pairs must be terminated with 120 Ω at the far end). In serial mode the driver can be bypassed (`MAIN_CFG.rs422 = 0`) for short/local runs — then the outputs are standard CMOS and termination resistors should not be used. App hint: unterminated RS422 drivers cannot handle pull-ups/pull-downs and give varying edge shapes. The **only** place an external high-current driver is needed is Encoder Link activation, where external drivers must overcome the chip's outputs: source/sink **≥ 40 mA** and drive A+/A– to **< 0.8 V and > 2.4 V**.
- **UVW/LED sharing note**: in the ABZUVW mode the three pairs are split into single-ended A+/B+/Z+ and U–/V–/W– (UVW appears inverted; set `uvwpol` for normal polarity).

6.2 UVW commutation output

- **Pole pairs supported: 1 … 32 pole pairs (= 2 … 64 poles).** `UVW_CFG.pairs` (0x0100 bits 4:0): value **0 = 32 UVW cycles per input cycle**, values 1…31 = that many UVW cycles per input cycle. The features/description text phrase this as "2...64 poles", "1 to 32 pole-pair motors", "up to 64 poles (32 pole pairs)".
- **Commutation type: block (3-phase square-wave) UVW signals**, "for commutating brushless motors". A sine-commutation output is **not** provided (not mentioned anywhere in the text). `OUTPUT.uvwen` (bit 7) enables the UVW outputs.
- **UVW position resolution / phase programming**

- UVW phase shift relative to the sin/cos input cycle: `UVWPS = 4 × UVWPH.msb + PHASE_LSB.uvw`, `uvwph[°] = UVWPS × 360° / 1024` — i.e. **10-bit phase resolution across the input cycle** (`UVWPH` 0x010B = UVWPH[9:2], `PHASE_LSB.uvw` 0x010C bits 3:2 = UVWPH[1:0]).
- `OUTPUT.uvwpol` (bit 4) inverts the UVW polarity and **shifts UVW phase by 180 °**.
- `OUTPUT.uvwdir` (bit 5) reverses the phase sequence (rotation direction).
- `UVW.hyst` 0…31 for standstill jitter reduction (`uvwhyst = ± code × 360°/2048`).
- UVW frequency limit flag: `STAT_VAL.inclim` also indicates UVW output frequency > **8.33 MHz**.
- **Timing/alignment rules**: programmable UVW phase correction aligns UVW to the motor/input cycle; the exact electrical sector boundaries per UVW cycle are **not determinable from text** (Figures show 2-pole UVW, but sector widths are not stated numerically).
- **How it is programmed**: `UVW_CFG.pairs`, `UVWPH`, `PHASE_LSB.uvw`, `UVW`, `OUTPUT.uvwen/uvwpol/uvwdir`, all EEPROM-stored (E type). Pin configuration mode cannot produce UVW (`UVW_CFG = 0`, "UVW disabled").

6.3 BiSS output

Not present on the iC-TW28. The only occurrence of "BiSS" in the entire datasheet is packaging footnote 6: "iC-TW29 uses pin 13 as GPIO, and pin 32 as BISSEN (high selects BiSS)." That is a **compatibility note about a different device (iC-TW29)**. There is no BiSS-B/BiSS-C description, no BiSS clock rate, no BiSS frame structure and no BiSS register access anywhere in the iC-TW28 datasheet text. → BiSS-C/BiSS-B support, clock rates, frame lengths, timing and master hints: **not determinable from text (not a feature of this chip)**.

6.4 Other serial interfaces

SPI (slave, 4-wire) — available in serial configuration mode only (PINCFG = DVSS). - Mode **CPOL = 0, CPHA = 0** only: SCLK rests low, SI sampled on SCLK rising edge, SO changes on the falling edge. - Max SCLK **20 MHz** (`f(SCLK)` item 704; note that condition is "PINCFG connected to DVSS"), min cycle time `tC1` = 50 ns. - All packets **32 bits, MSB first**; **overlapped** structure: the response to command *n* is returned while command *n+1* is shifted in. The very first response (to packet 1) is undefined. - xSS must be high when xRST is released. xSS must not be pulsed without exactly 32 SCLK edges. - **Command packet** bits 31:24 = SPI control byte: bit31 `st` (must be 1), bits30:29 `rm` (read mode), bits28:27 `wm` (write mode), bits26:24 reserved (0). - Write modes `wm`: 0 = Null write (read only), 1 = Multi-Cycle Counter Write, 2 = Multi-Cycle Counter Atomic Read/Write, 3 = Register Write (bits 23:8 = address, bits 7:0 = data). - Read modes `rm`: 0 = Position and Status Read (20 ns or 320 ns update), 1 = Captured Position and Status Read, 2 = Sin/Cos/Zero ADC Read (320 ns), 3 = Register Data and Position Read (bits 23:8 = register address, wm must be 0). - **Response packets** - `rm = 0`: bits 31:24 SPI status byte (`ffl t` fatal fault, `irq` interrupt active, `zcl` zero capture lost, `zc` zero capture occurred, `mcrl` counter rollover lost, `mcr` counter rollover occurred), bits 23:10 multi-cycle counter (0…16,383), bits 9:0 angle (0…1023). - `rm = 1`: status byte, bits 23:10 captured multi-cycle counter, bits 9:0 captured angle. - `rm = 2`: bits 31:28 reserved, 27:18 corrected Sin ADC (signed 10-bit), 17:8 corrected Cos ADC (signed 10-bit), 7:0 corrected Zero ADC (signed 8-bit). - `rm = 3`: bits 31:24 register data, 23:10 counter, 9:0 angle. - **Timing** (items I001–I011): tC1 ≥ 50 ns; clock low ≥ 15 ns; clock high ≥ 15 ns; tS1 (xSS low before SCLK rise) ≥ 80 ns; tH1 ≥ 50 ns; tW1 (between xSS high→low) ≥ 200 ns (≥ 600 ns if the preceding command was an ADC read); tS2 (SI stable before SCLK rise) ≥ 5 ns; tH2 ≥ 10 ns; tP1 (SO valid after xSS low) ≤ 60 ns; tP2 (SO Hi-Z after xSS high) ≤ 25

ns; tP3 (SO valid after SCLK fall) ≤ 20 ns. - Bussing: shared SCLK/SI/SO, individual xSS, push-pull xIRQ recommended. - Chaining: shared SCLK/xSS, SI→SO daisy chain, 3 × 32 = 96 bits for three devices, commands execute together when xSS goes high; xIRQ must be open-drain and wire-OR'd with a pull-up.

Encoder Link (2-wire, SPI-like) — serial configuration mode only; uses **A+ (clock input) and A– (data in/out)** - Enables field reconfiguration/diagnostics; **cannot** read angle, multi-cycle counter or captured position. - Activation sequence (Figure 24): external drivers over-power A+ and A– (both high for t1 = 0.25…2 ms, then the sequence t2/t3/t4 each = t1 ±10 %); total activation 1…8 ms; afterwards A+/A– are both low and the interface is active. - Timing (items I105–I110): ELink clock **max 1000 kHz** (must be ≤ 500 kHz for operation above 85 °C); clock high ≥ 200 ns; clock low ≥ 200 ns; input setup ≥ 200 ns; input hold ≥ 200 ns; output propagation ≤ 200 ns. - Write command = **32 bits**: ELink control byte (bit31 `st` = 1, bit30 `wr` = 1, bits 29:24 reserved), bits 23:8 register address, bits 7:0 data. Read command = **40 bits**: command 32 bits + 8-bit wait byte, during which the host releases A– and the iC-TW28 drives the addressed register byte back (sampled on the A+ rising edge). - Deactivation: power cycle or xRST toggle. Can be disabled with `MAIN_CFG.elinkoff = 1` (recommended in SPI-Only and ABZUVW modes to avoid accidental activation).

PWM / other outputs - **LED PWM output** (pin 26): delta-sigma modulated, `LED_CFG.freq` selects DSM 3.125 / 6.25 / 12.5 / 25 MHz ⇒ PWM 12.2 / 24.4 / 48.8 / **97.6 kHz**; 8-bit duty (`LED_START` 0…255 initial, `LED_PWM` 0x0010 actual); polarity, open-drain/push-pull, loop buffer 8/16/24/32; intensity control moves the PWM value ±1 at 500 Hz. - **xIRQ** digital fault/interrupt output (push-pull or open-drain). - **No I²C, no SSI, no analog output, no sine output** are present in the datasheet.

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7. Register map

Legend (Type column in the datasheet): **E** = stored in EEPROM and restored at startup/restart; **D** = dynamic, may be modified by the chip (user may also write it); **R** = dynamic, read-only. Registers without a code are user read/write. Registers not listed are reserved.

7.1 Full register map (addresses and bit fields)

Address	Name	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	Type
0x0000	`MAIN_CFG`	0	0	elinkoff	irqpp	rs422	noise	input[1:0]		E
0x0001	`LED_CFG`	buffer[1:0] (b7:6)		freq[1:0] (b5:4)		auto	odrain	pol	en	E
0x0002	`LED_START`	Initial (starting) LED PWM value (0…255)								E
0x000B	`TEST`	0	0	0	0	0	z	0	we	—
0x0010	`LED_PWM`	Actual LED PWM value								R

Address	Name	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	Type
0x0100	UVW_CFG	0	0	0	pairs[4:0]					E
0x0101	INTER0	interlsb (INTER[7:0])								E
0x0102	INTER1	0	0	0	div[2:0]			intermsbs[1:0]		E
0x0103	AB	0	0	0	hyst[4:0]					E
0x0104	UVW	0	0	0	hyst[4:0]					E
0x0105	FALARM	input frequency alarm level (1…128)								E
0x0106	ABLIMIT	AB frequency limit (0…255)								E
0x0107	ZERO0	mode	clr	threshold[5:0]						E
0x0108	ZERO1	0	0	0	0	zwidth[3:0]				E
0x0109	OUTPUT	uvwen	abzen	uvwdir	uvwpol	abzdir	zpol	bpol	apol	E
0x010A	ZPHASE	ZPHASE[9:2] (msb)								E, D
0x010B	UVWPH	UVWPH[9:2] (msb)								E
0x010C	PHASE_LSB	0	0	0	0	uvw[1:0]		z[1:0]		E, D
0x020C	Z_ADC	Zero ADC value (8-bit)								D
0x021A	S_ADC	Sine ADC MSB								D
0x021C	C_ADC	Cosine ADC MSB								D
0x0300	FILT_CFG	0	0	0	kpmax[2:0]			fb	auto	E
0x0301	FILT_LAG	0	0	0	threshold[4:0]					E

Address	Name	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	Type
0x0302	FILT_K	0	0	ki[1:0]		kp[2:0]				E, D
0x0400	STAT_CFG	0	0	enz	enc	long	filter[2:0]			E
0x0401	STAT_SEL	res	adapt	scamp	lagfatl	inclim	laglim	falarm	oflow	E
0x0402	STAT_IE	res	adapt	scamp	lagfatl	inclim	laglim	falarm	oflow	E
0x0403	STAT_HIZ	res	adapt	scamp	lagfatl	inclim	laglim	falarm	oflow	E
0x0404	STAT_VAL	res	adapt	scamp	lagfatl	inclim	laglim	falarm	oflow	D
0x0405	STAT_LATCH	res	adapt	scamp	lagfatl	inclim	laglim	falarm	oflow	D
0x0406	STAT_FATAL	0	0	0	0	0	interr	ee2bit	eechk	D
0x0601	EE_ADDR	EEPROM address to read or write								—
0x0602	EE_DATA	EEPROM data								D
0x0603	EE_STAT	Validity of read EEPROM data								D
0x4000	COMMAND	Command register (see 7.3)								D
0x4001	START	0	0	nostart	mode[1:0]		wait[2:0]			E
0x4002	ADAPT_CFG0	zphase	zgain	zofs	scgain	scofsa	scph	scbal	scofs	E
0x4003	ADAPT_CFG1	0	0	xcalee	zcal	stop	p[2:0]			E
0x4004	SC_AMP_TARG	Sin/Cos amplitude monitor target (**recommended 150**, 0…180)								E
0x4005	SC_AMP_LOW	Amplitude monitor low limit (0…SC_AMP_TARG)								E
0x4006	SC_AMP_HIGH	Amplitude monitor high limit (SC_AMP_TARG…180)								E

Address	Name	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	Type
0x4007	S_OFS_BASE	Sin channel digital offset base (±127)								E
0x4008	C_OFS_BASE	Cos channel digital offset base (±127)								E
0x4009	SC_OFS_LIM	S/C digital offset limit (0…127)								E
0x400A	SC_OFS_TH	S/C offset residue threshold (0…127)								E
0x400B	SC_BAL_BASE	Sin/Cos balance base (±127)								E
0x400C	SC_BAL_LIM	Sin/Cos balance limit (0…127)								E
0x400D	SC_BAL_TH	Sin/Cos balance residue threshold (0…127)								E
0x400E	SC_PH_BASE	Sin/Cos phase base (±127)								E
0x400F	SC_PH_LIM	Sin/Cos phase limit (0…127)								E
0x4010	SC_PH_TH	Sin/Cos phase residue (0…127)								E
0x4011	Z_PH_TH	Z channel phase residue (0…127)								E
0x4012	S_OFS_COR	Sin digital offset correction (±127)								E, D
0x4013	S_OFSA_COR	Sin analog offset correction (±31)								E, D
0x4014	C_OFS_COR	Cos digital offset correction (±127)								E, D
0x4015	C_OFSA_COR	Cos analog offset correction (±31)								E, D
0x4016	SC_BAL_COR	Sin/Cos balance correction (±127)								E, D
0x4017	SC_GN_COR	Sin/Cos digital gain correction (0…255)								E, D
0x4018	SC_GNA_COR	Sin/Cos analog gain correction (0…23)								E, D
0x4019	SC_PH_COR	Sin/Cos phase correction (±127)								E, D

Address	Name	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	Type
0x401A	`Z_OFSA_COR`	Z analog offset correction (±31)								E, D
0x401B	`Z_GNA_COR`	Z analog gain correction (0…23)								E, D
0x401C	`WATCHDOG`	Watchdog: should read 31 (0x1F) when OK								D
0x4020	`SC_AMP`	Sin/Cos amplitude √(sin²+cos²) (0…255), updated every 500 µs								R
0x4021	`S_AMP`	Sin channel amplitude (0…255), updated once per input cycle or less								R
0x4022	`C_AMP`	Cos channel amplitude (0…255)								R
0x4023	`S_OFS_RES`	Sin offset residue (±127)								R
0x4024	`C_OFS_RES`	Cos offset residue (±127)								R
0x4025	`SC_BAL_RES`	Sin/Cos balance residue (±127)								R
0x4026	`SC_PH_RES`	Sin/Cos phase residue (±127)								R
0x4028	`Z_PH_RES`	Z phase residue (±127)								R
0xE000	`DEVICE_ID`	0x0C								R
0xE002	`REV0`	Revision LSB								R
0xE003	`REV1`	Revision MSB								R

Reset/default values: the datasheet does not print a numeric power-on default column for the register map; the only documented "defaults" are (a) the **pin-configuration-mode values** (Tables 6 and 7, below) and (b) the factory default noted for `START.wait = 3`. `TEST` (0x000B) is explicitly stated to be 0 at every startup. **Per-register power-on defaults for serial mode: not determinable from text.**

7.2 Pin-configuration-mode register values (Tables 6 and 7 — these are also useful as a safe starting set)

Note: pin-configured register values are **not** stored in EEPROM.

Register.bit	Value	Meaning
`MAIN_CFG.input`	from C3	input range
`MAIN_CFG.filter` (noise)	0	500 kHz max input freq., 1.5 µs latency
`MAIN_CFG.rs422`	1	RS422 ABZ outputs
`MAIN_CFG.irqpp`	0	open-drain xIRQ
`MAIN_CFG.elinkoff`	1	Encoder Link not available
`LED_CFG`	0	LED PWM disabled
`UVW_CFG`	0	UVW disabled
`INTER`	from C0/C3	interpolation
`INTER1.div`	0	no post-AB divider
`AB.hyst`	from C1	AB hysteresis/filtering
`UVW.hyst`	0	0 ° UVW hysteresis
`FALARM`	0	no input frequency alarm
`ABLIMIT`	from C2	AB frequency limit
`ZERO0.threshold`	10	42 % of signal amplitude
`ZERO0.mode`	0	position capture on Z
`ZERO0.clr`	0	never clear multi-cycle counter
`ZERO1.zwidth`	0	1 AB edge (quadrature state)
`OUTPUT.apol/.bpol`	0	normal polarity
`OUTPUT.zpol`	0	Z active high
`OUTPUT.abzdir/.uvwpol/.uvwdir`	0	normal direction/polarity/rotation
`OUTPUT.abzen`	1	ABZ outputs
`OUTPUT.uvwen`	0	no UVW outputs
`ZPHASE`	0	affected by C3
`UVWPH`	0	no UVW phase shift
`FILT_CFG.auto/.fb`	0 / 0	static kp, no feedback delay
`FILT_CFG.kpmax`	6	max. dynamic kp limit
`FILT_LAG.threshold`	12	filter lag threshold

Register.bit	Value	Meaning
`FILT_K.kp`	from C1	filter P coefficient
`FILT_K.ki`	3	maximum ki (recommended)
`STAT_CFG.filter`	3	2.5 ms status filter
`STAT_CFG.long`	1	IRQ extended by 40 ms
`STAT_CFG.enc/.enz`	0 / 0	no IRQ on counter overflow / position capture
`STAT_SEL`	0	STAT_VAL used for IRQ
`STAT_IE.oflow/.falarm/.laglim/.adapt/.res`	0	no IRQ
`STAT_IE.inclim/.lagfatl/.scamp`	1	IRQ on AB/UVW limit, fatal lag, amplitude lo/hi
`STAT_HIZ`	0	outputs always enabled
`START.wait`	3	10 ms startup wait (factory default)
`START.mode`	1	same-phase startup (always used in pin mode)
`START.nostart`	0	automatic startup
`ADAPT_CFG0`	from C2	adaption channels
`ADAPT_CFG1.p`	1	slow auto-adaption rate (1/16)
`ADAPT_CFG1.stop`	1	stop auto-adaption on SCAMP
`ADAPT_CFG1.zcal`	from C3	Z calibration with xCALIB
`ADAPT_CFG1.xcalee`	1	store to EEPROM when xCALIB de-activates
`SC_AMP_TARG`	150	sin/cos amplitude target
`SC_AMP_LOW`	90	−40 % of target
`SC_AMP_HIGH`	180	+20 % of target
`S_OFS_BASE`, `C_OFS_BASE`, `SC_OFS_LIM`, `SC_OFS_TH`, `SC_BAL_BASE`, `SC_BAL_LIM`, `SC_BAL_TH`, `SC_PH_BASE`, `SC_PH_LIM`, `SC_PH_TH`, `Z_PH_TH`	0 / 127 as listed	not used in pin mode
`Z_GNA_COR`	8	affected by C3

7.3 COMMAND register (0x4000)

The register auto-clears to 0x00 when the command has executed.

Value	Command
0x00	Ready/idle
0x01	Start/restart interpolation (with currently loaded values)
0x02	Stop interpolation (ABZ/UVW → Hi-Z, LED deactivated, xIRQ still active)
0x10	Load configuration and COR registers from EEPROM (does not start)
0x11	Write configuration + COR + LED values to EEPROM (may take up to **1 s**)
0x12	Copy COR values to BASE registers (S/C offset, balance, phase)
0x13	Read EEPROM address (needs ≥ **1 ms**; result in EE_DATA, validity in EE_STAT)
0x14	Write EEPROM address (needs ≥ **20 ms**)
0x16	Load configuration and COR registers from EEPROM **and start** interpolation
0x17	Write COR register values to EEPROM
0x20	Stop auto calibration
0x21	Auto calibrate sin and cos parameters
0x22	Auto calibrate Z channel parameters
0x23	Auto calibrate all parameters
others	Reserved

Commands marked with * in the datasheet (0x11, 0x14, 0x17) do nothing while the EEPROM is locked (`TEST.we = 0`). EEPROM map: 0x00–0x03 device serial number (read-only, factory), 0x04–0x3B reserved, **0x3C–0x3F four user data bytes**, 0x40–0xFF reserved. `EE_STAT`: 0 = valid, 1 = single-bit error corrected (data valid), 2 = double-bit error (data invalid).

7.4 Pin-configuration-mode tables (hardware-only configuration)

C0 — interpolation factor (C3 selects group I0/I1)

C0 level	Group I0	Group I1
11	256	250
10	128	200
9	64	180
8	32	125
7	16	100
6	12	80
5	8	50
4	6	40
3	4	25
2	3	20
1	2	10

C0 level	Group I0	Group I1
0	1	5

C1 — hysteresis / filtering: levels 11–8 = ±2.8 / ±1.4 / ±0.7 / ±0.35 ° with **heavy** filtering; 7–4 = same four values with **medium**; 3–0 = same four values with **light**.

C2 — AB frequency limit & auto-adaption: levels 11–6 = 195 kHz/1.28 µs, 390 kHz/640 ns, 781 kHz/320 ns, 1.56 MHz/160 ns, 3.12 MHz/80 ns, 6.25 MHz/40 ns with adaption **on**; levels 5–0 = same frequencies with adaption **off**.

C3 — input range, interpolation group, auto Z calibration: levels 10/9 = I1 with High/Low range, auto Z yes; 7/6 = I0 High/Low, auto Z yes; 4/3 = I1 High/Low, auto Z no; 1/0 = I0 High/Low, auto Z no; 11, 8, 5, 2 reserved.

Configuration resistor table (Table 1) — nominal level → Rx1/Rx2 (kΩ): 11 → 3.30 V (0.00 / ∞); 10 → 3.00 (10.0 / 100); 9 → 2.70 (11.5 / 51.1); 8 → 2.40 (11.5 / 30.1); 7 → 2.10 (16.2 / 28.7); 6 → 1.80 (9.53 / 11.5); 5 → 1.50 (11.5 / 9.53); 4 → 1.20 (28.7 / 16.2); 3 → 0.90 (30.1 / 11.5); 2 → 0.60 (51.1 / 11.5); 1 → 0.30 (100 / 10.0); 0 → 0.00 (∞ / 0.00).

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8. Timing, reset, power-on behaviour

Item	Value
POR turn-on threshold (DVDD rising, xRST tied to DVDD)	**2.5 / 2.7 / 3.0 V** (min/typ/max)
Undervoltage reset threshold (DVDD falling)	2.2 / 2.4 V
Startup time to AB output function	**4.5 ms** with `START.wait = 0`; **14.5 ms** with `START.wait = 3` (factory default)
Startup wait times	0 / 1 / 3 / 10 / 30 / 100 / 300 ms (`START.wait`; serial-configurable total wait 0…514 ms)
xRST RC recommendation	1 kΩ + 100 nF ≈ **100 µs** delay; larger R → proportionally longer
VDD rise time if xRST is tied directly to VDD	**≥ 40 µs** from 0 V to 3.3 V (release W/W1 notes)
xCALIB at power-on	xCALIB must rise **with** the supply — must not be held low by a capacitor
Startup sequence	POR monitors supply until 2.7…2.9 V → (pin mode) wait 10 ms → read configuration from EEPROM → start ABZ/UVW generation; (serial mode) programmable wait, and startup may be inhibited (`START.nostart`)

Item	Value
On error at startup	Outputs are not enabled; device goes to an idle state with xIRQ asserted
Startup modes (`START.mode`)	0 = relative (A/B state identical after every startup), 1 = same phase (A/B state a function of sensor angle), 2 = absolute burst (angle counted out on A/B at startup, max **2 × inter** edges ±180 ° at the ABLIMIT rate), 3 reserved. Pin mode always uses same-phase. Before startup the AB outputs are Hi-Z
Same-phase caveat	Only use if `INTER1.div` ∈ {0,1,3,7} (divider off or power of 2) and inter is an integer
Latency	1.5 µs (`MAIN_CFG.noise = 0`) / 1.9 µs (`MAIN_CFG.noise = 1`) from input change to AB/UVW output
Angle/MCC internal update	320 ns with ABZ enabled; 20 ns with ABZ disabled (valid only for INTER = 0)
ADC sample/update	corrected sin/cos/zero values updated every **320 ns**; `S_ADC` / `C_ADC` / `Z_ADC` registers update irregularly at ≈50 µs
`SC_AMP` update	every 500 µs
`S_AMP` / `C_AMP` update	once per input cycle, or less at high speed
AB minimum transition distance `tMTD`	1/f_osc ≈ 20 ns at `ABLIMIT = 0x00`
RS422 timing specs	`tAB` (phase A vs B) = 25 %, `twhi` (duty at A/B) = 50 %
Digital output edges	t_r / t_f = 20 ns (DVDD = 3.3 V, C_L = 50 pF, 10–90 %)
SPI timing	see §6.4
Encoder Link timing	see §6.4
Jitter example (SPI-Only, 500 kHz input, 2 µs SPI period)	position uncertainty from the 20 ns update = **0.01 input cycles = 3.6 ° = 1 %**; for the ADC read bank the 320 ns update gives 0.16 cycles = 57.6 ° = 16 %
Oscillator	48…51 MHz at 3.1 V, 49…52 MHz at 3.6 V (typ. 50 MHz), tempco 225 ppm/K
Frequency formula (pin mode example)	`f_input = AB_frequency / interpolation`; e.g. 1.56 MHz / 250 = 6.24 kHz
Lag/lag-fault behaviour	lag > 45 ° → `laglim` (filter disabled, outputs valid); lag ≥ 180 ° → `lagfatl` (fatal, outputs invalid)

9. Electrical characteristics

Absolute maximum ratings

Item	Value
VDD at DVDD/AVDD/IOVDD (to their grounds)	−0.3 … **4.1 V**
Pin voltage at any pin	−0.3 … AVDD + 0.3 V
Input current into any pin	−2 … +2 mA
ESD susceptibility	**HBM 4 kV** (100 pF through 1.5 kΩ)
Junction temperature	−40 … 150 °C
Storage temperature	−40 … 150 °C
Latch-up rating	**not determinable from text** (no latch-up specification is given)

Thermal / operating

Item	Value
Operating ambient temperature T_a	**−40 … +125 °C**
Thermal resistance R_thja (QFN32 on JEDEC 51 PCB)	**40 K/W**
Operating supply	AVDD = DVDD = IOVDD = **3.1 … 3.6 V**

Current consumption (3.3 V, f_in = 1 kHz, inter = ×256, ABZ and UVW active)

Rail	Current
I_AVDD	**15 mA**
I_DVDD	**22 mA**
I_IOVDD	**85 mA** with RS422 drivers enabled and 120 Ω-terminated quadrature outputs; **2.5 mA** with outputs open
IOVDD loading note	"IOVDD can require up to 100 mA depending on loads"

Other key limits

- RS422 drivers: nominal **18 / 27 mA** into 120 Ω (item 901, D3 revision set a min limit); short-circuit high ≤ −55 mA, low ≥ 35 mA (release W note: these limits can be exceeded at low/high temperatures).
- Digital outputs: ±10 mA per pin indefinite; V_s(hi) / V_s(lo) ≈ 0.7 V at ∓4 mA; short-circuit −30/−16 mA and 16/30 mA.
- LED output: ±15 mA continuous; V_out(hi) 2.7 V at −10 mA; V_s(hi) 1 V at −10 mA; V_s(lo) 1 V at 10 mA; short-circuit −40 / +40 mA.
- Bias outputs: **VC = 50 % AVDD**; **VREF = VC − 1.0 V** (dVREF −1.1 / −1.0 / −0.9 V).
- Internal EEPROM: write cycles **7 / 50** for chip releases W and W1 and **1000** for release V1 (columns min/typ/max in the extraction); design-review note for **W1 limits the number of write**

cycles to 16. Write temperature range −40…+85 °C; read temperature range −40…+125 °C; raw data retention **10 years**.
- EEPROM error protection: checksum plus single-bit error correction and double-bit error detection.

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10. What an FPGA must reproduce

Checklist of every iC-TW28 function a designer would have to re-implement to build a functionally equivalent sin/cos interpolating encoder. **D** = digital-friendly (fits in FPGA fabric); **A** = analog, must be done with external components (ADC front end, PGA, references, line drivers, EEPROM); **H** = hybrid (digital core with an analog or mixed-signal support requirement).

#	Function (datasheet basis)	D/A/H	Notes for FPGA implementation
1	Differential sin/cos input amplification, 20 mV…700 mV (low) or 65 mV…2 V (high) peak	**A**	External instrumentation/PGA stage or external ADC with adequate range; the chip's 6…40.5 dB / 1.5 dB-step analog gain block has no digital equivalent
2	Configurable 9 dB input attenuation for rail-to-rail (GMR/TMR/optical) sensors	**A**	Resistive attenuator + PGA
3	Input noise filter after PGA (2 settings)	**A** (or H)	Analog anti-alias filter; a digital decimation filter is an alternative but changes latency
4	Analog offset/gain correction (coarse, pre-ADC)	**A**	6…40.5 dB in 1.5 dB steps; offset DAC into the PGA. In an all-digital design this must be an analog trim or a higher-resolution ADC with digital trim
5	10-bit sin/cos ADCs (up to ~700 kHz input ⇒ ≥ several MS/s each)	**A**	External dual ADC; note the chip's ADC "sample rate" is **not stated** in the datasheet — equivalent update observable is the 320 ns corrected-ADC update
6	8-bit zero/index ADC + comparator	**A** (+D)	Analog ADC + digital comparator with programmable threshold (see #19)
7	Digital offset correction per channel (`S_OFS_COR`, `C_OFS_COR`, ±127; step 3.9 mV/gain)	**D**	Simple adder with signed coefficient; auto-calibration loop is a digital accumulator
8	Digital amplitude-balance / gain equalisation (`SC_BAL_COR` ±127, `SC_GN_COR` 0…255; step 0.2 %)	**D**	Multiplier by (1 ± β/2) with β from the coefficient
9	Digital quadrature/phase correction (`SC_PH_COR` ±127; ±26 °, step 0.22 °)	**D**	Cross-coupling matrix (sin' = sin + k·cos) or CORDIC pre-rotation

#	Function (datasheet basis)	D/A/H	Notes for FPGA implementation
10	Auto-calibration: offset, gain, balance, phase (+Z channel), driven by `xCALIB` or SPI commands 0x21/0x22/0x23, terminated/stored by 0x20/0x11/0x12/0x17	**D**	Least-squares / servo estimator over a few hundred input cycles; the chip's exact estimator algorithm and convergence time are **not determinable from text**
11	Auto-adaption during operation, per-channel enables (`ADAPT_CFG0`), rate `ADAPT_CFG1.p` (1/32, 1/16, 1/8, 1/4, 1/2, or 1 COR step per input period), stop-on-`scamp`, Z-cal and store-on-xCALIB flags	**D**	Per-coefficient up/down counters/accumulators clocked once per input cycle
12	Residual calculation (offset, balance, phase; RES registers) and thresholds (`SC_*_TH`, `Z_PH_TH`)	**D**	Accumulate residual error per cycle, compare with threshold, set status bit
13	Adaption deviation detection (COR vs BASE vs LIM)	**D**	Absolute difference comparators against EEPROM-stored base/limit pairs
14	`arctan(Sin/Cos)` **CORDIC**, 10-bit output, one angle per input cycle	**D**	Classic pipelined/iterative CORDIC; 10-bit output = 1024 increments/cycle. Iteration count / pipeline depth of the original is not stated
15	Interpolation scaling: `INTER = 256·INTER1.msbs + INTER0.lsb`; inter = 256 for INTER = 0, else inter = INTER/4 (2…256 + fractional values)	**D**	Phase accumulator / angle scaler; carry-in/out increments the multi-cycle counter
16	Optional **post-AB divider** div = `INTER1.div` + 1 = 1…8 after the ABZ generator, with Z bypassing the divider	**D**	Counter-based divider; remember Z is not divided and Z-to-AB sync may be lost
17	Digital filter: PI servo loop (kp 0…6, ki 0…3, static/dynamic kp, `FILT_CFG.kpmax` 3…6, lag threshold 1…31, optional **640 ns** feedback delay)	**D**	A 640 ns delay at 50 MHz is 32 cycles — the datasheet's delay is stated as an analog-ish figure; implementable as a delay line. Instability rules must be honoured (kp = 0 with ki = 1 is unstable; ki = 0 ⇒ kp = 0; fb = 1 ⇒ kp ≥ 3)
18	Hysteresis in the AB and UVW paths (`AB.hyst`, `UVW.hyst` 0…31; ±code × 360°/2048)	**D**	Digital window comparator on the position value
19	Z/index channel: comparator threshold `ZERO0.threshold` (±31), gating window = input > 4×threshold, Z pulse width `ZERO1.zwidth` (0…15, `(zwidth+1)/div` AB edges, must be < INTER), Z phase `ZPH = 4·ZPHASE.msb + PHASE_LSB.z` (`×360°/1024`), pulse centred in the window by auto calibration, gate-width rule `zwidth° < gatewidth° < 720° − zwidth°`	**D** (+A for the zero sensor)	Digital comparator on the (externally) digitized index signal; Z pulse generation must be synchronous to the AB outputs

#	Function (datasheet basis)	D/A/H	Notes for FPGA implementation
20	Z test mode (`TEST.z`) for observing the gating window on B outputs	**D**	Diagnostic mux — useful for bring-up
21	ABZ output generation: quadrature A/B, 4 edges per output cycle, programmable polarity (`apol`, `bpol`), direction (`abzdir`), Z polarity (`zpol`), 25 % A-to-B phase relationship, 50 % duty	**D**	Straightforward quadrature state machine
22	AB output frequency limiter: `fab = 12.5 MHz/((ABLIMIT+1)·div)`, `t_edge = 20·(ABLIMIT+1)·div` ns, 0…255 settings, applies before the divider; lag catch-up then fatal-lag behaviour	**D**	Programmable edge-spacing timer + lag accumulator + fatal fault at 180 °
23	UVW block commutation: `UVW_CFG.pairs` 1…32 pole pairs (0 = 32), 10-bit phase shift `UVWPS = 4·UVWPH.msb + PHASE_LSB.uvw` (`×360°/1024`), `uvwpol` (180 ° shift), `uvwdir`, `UVW.hyst` 0…31, UVW > 8.33 MHz flag	**D**	3-phase sector/block generator from the angle accumulator and pole-pair setting; sine commutation is **not** part of the original
24	SPI slave: 4-wire, **CPOL 0 / CPHA 0**, ≤ 20 MHz, 32-bit MSB-first packets, **overlapped** (response to command n returned during command n+1), control byte (st/rm/wm), read modes 0–3, write modes 0–3, status byte, register/address access	**D**	Must reproduce the overlapped packet timing exactly to be command-compatible; xSS must always be accompanied by exactly 32 clocks
25	Multi-cycle counter: 14-bit, 0…16,383; clear-on-Z option (`ZERO0.clr`); preset via SPI (wm = 1 / wm = 2); `mcs` sector-synchronization bit with ±1 correction rule; rollover flags at multiples of **4,096** with 4,096-cycle hysteresis (`mcr`, `mcrl`)	**D**	Sign/magnitude tracking from angle wrap-around; must handle direction reversal identically
26	Position capture register: 24-bit (14-bit MCC + 10-bit angle) on Z output rising edge or Z gating window (`ZERO0.mode`), `zc` / `zcl` status bits, capture IRQ via `STAT_CFG.enz`	**D**	Simple shadow register with overrun flag
27	Status & fault logic: 8 conditions (`oflow`, `falarm`, `laglim`, `inclim`, `lagfatl`, `scamp`, `adapt`, `res`), live `STAT_VAL`, latched `STAT_LATCH` (write-0-to-clear), `STAT_SEL` source select, `STAT_IE` IRQ enables, `STAT_HIZ` output disable, `STAT_FATAL` (eechk, ee2bit, interr) → fatal shutdown	**D**	Table-driven flag logic; honour the 0/10 µs/150 µs/2.5 ms/40 ms `STAT_CFG.filter` and the 40 ms IRQ extension
28	Amplitude monitor `√(sin²+cos²)`: `SC_AMP`, `SC_AMP_TARG` (recommended 150, max 180), `SC_AMP_LOW` / `SC_AMP_HIGH`, `SC_AMP` updated every 500 µs, `S_AMP` / `C_AMP` once per input cycle	**D**	Cordic/multiplier or
29	Input frequency alarm: `f[MHz] = 1.56·FALARM/256`, FALARM 1…128	**D**	Period counter on the angle wrap

#	Function (datasheet basis)	D/A/H	Notes for FPGA implementation
30	LED intensity control: 8-bit DSM/PWM at 12.2/24.4/48.8/97.6 kHz, ±1 step at 500 Hz, buffer 8/16/24/32, polarity and open-drain/push-pull, formula `LED_START = INT(I_nom/I_max · 256)`	**H**	Digital PWM/DSM is FPGA-friendly; the LED current stage (open-drain or external FET, R/C filter sizing) is external
31	Startup behaviour: `START.wait` (0/1/3/10/30/100/300 ms), `START.nostart`, three startup modes (relative / same phase / absolute burst with up to 2·inter edges), same-phase restrictions on `INTER1.div`, Hi-Z outputs before start	**D**	Needs an equivalent power-on state machine, plus EEPROM load before output enable
32	Reset / power-on: POR at 2.5…3.0 V, undervoltage at 2.2…2.4 V, external xRST, start-up in 4.5 ms (wait 0) or 14.5 ms (wait 3)	**H**	External supervisor/POR IC + FPGA reset sequencer
33	Internal 50 MHz oscillator (48…52 MHz, 225 ppm/K) usable as the timing reference	**A**	External crystal/oscillator; the FPGA should not rely on internal RC accuracy
34	EEPROM for configuration + calibration, with checksum, single-bit correction, double-bit detection, write cycle limit (≈16 for W1), 4 user bytes, device serial number	**A**	External I²C/SPI EEPROM or FPGA configuration flash; must implement the same protection class
35	RS422 differential ABZ/UVW line driver (18…27 mA, 120 Ω terminated, or bypassed to CMOS)	**A**	External RS422 transceiver(s) — the chip integrates this, an FPGA cannot
36	BiSS / SSI / absolute serial output	—	**Does not exist on the iC-TW28**; if the target application needs BiSS-C the FPGA must implement the whole protocol itself (see iC-TW29 for the BiSS-capable sibling part)
37	Encoder Link 2-wire field-reconfiguration interface (A+/A–, ≤ 1 MHz, activation by over-driving the outputs with ≥ 40 mA drivers, 32-bit write / 40-bit read frames)	**D** (+A driver)	Digital protocol is reproducible; the external over-drive drivers are analog/drive-strength work. Note that on the original, Encoder Link cannot read position
38	Watchdog register (must read 31 = 0x1F within ≥ 1 ms of being cleared)	**D**	Trivial
39	Electrical envelope: 3.1…3.6 V, ~15/22/85 mA rails, −40…+125 °C ambient, 40 K/W, 4 kV HBM ESD	**A**	Board-level/analog responsibility

Minimum viable FPGA partition (summary): items **7–30, 31, 32 (digital part), 33 (as an input clock), 37, 38** are the digital core that must be reproduced for functional equivalence. Items **1–6, 30 (LED drive), 33 (oscillator), 34 (EEPROM), 35 (RS422), 39** are analog/board-level and must come from external devices — in particular the interpolator cannot function without an external dual ADC that meets or exceeds the original 10-bit sin/cos and 8-bit zero conversion, and without an external RS422 line driver if differential ABZ/UVW outputs are required.

Extraction caveats encountered (do not treat as datasheet facts)

1. The features list says "AB output frequency of up to 12.5 MHz" while the description says "up to 12 MHz (20 ns edge spacing)". The ABLIMIT formula uses **12.5 MHz**, consistent with a 20 ns minimum edge spacing at 4 edges/cycle.
2. Sin/cos ADC resolution: block-diagram text says 10-bit; the `S_ADC` / `C_ADC` register text says "MSB of the 12-bit sin channel ADC value". The two are not reconciled in the text.
3. Input impedance of the loss-detect and high ranges: elec. char. items 111 give 0.240 MΩ / 0.670 MΩ while Table 114 gives 220 kΩ / 620 kΩ.
4. EEPROM write-cycle limit: elec. char. item 501 gives 7 / 50 for releases W and W1 and 1000 for V1, while the W1 design-review note limits W1 to **16** cycles.
5. No power-on default column exists in the register map for serial configuration mode; only the pin-mode values (Tables 6/7) and `START.wait = 3` (factory default) are stated.
6. `START.mode = 1` (same phase) is described as "always used in pin configuration mode", and Table 6 lists `START.mode = 1` for pin mode.
7. The `UVWPH` phase formula is printed in the extraction as `uvwph[°] = uvwph × 360° / 1024`, which is self-referential; the variable computed by `UVWPS = 4 × UVWPH.msb + PHASE_LSB.uvw` is evidently the intended operand.